SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT (2 Questions)

rse Code:	ITA05	Course Title:	COMPUTER VISION
Assessed:	CO2 – Manipulation (BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
htage:	20% of CIA	Total Marks:	20 (2 Questions × 10 Marks)
Statement:	*Apply image enhancement and segmentation techniques for extracting meaningful regions from images.(BL3)*		
nt Name:	P.V.D.Nagendra Reddy	Reg. No.:	192472399
n / Batch:	Slot D / 2024	Date:	04/08/26.

of Conduct

gendra (192472399) (Name / Reg No) certify that this submission is my original work and that I have ed to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will n disciplinary action.

ure of the Student: NagendraReddy

tions

This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.

No negative marking. Unanswered questions carry zero marks.

No calculators or electronic devices permitted.

Duration: 60 minutes.

Annexure A – Scenario Based Assessment

Scenario 1: Manufacturing Quality Inspection

An automated quality inspection system captures images of metal components. Electrical interference introduces repetitive horizontal stripes across the images, making defect detection unreliable.

(a) Interpret the scenario and identify the type of noise present.
(b) Explain how this degradation affects inspection accuracy.
(c) Recommend an appropriate enhancement technique to remove the interference.
(d) Justify why your selected approach is more suitable than spatial-domain methods.
(e) Explain your answer in a logical sequence.

Scenario 2: Autonomous Vehicle Navigation

A self-driving car captures road images during heavy rain. Although noise is present, the lane markings and road edges must remain sharp for safe navigation.

(a) Interpret the scenario and explain the challenge involved.
(b) Identify the importance of preserving edges while removing noise.
(c) Recommend a suitable filtering technique.
(d) Justify why your selected filter preserves important image details.
(e) Present your explanation clearly and systematically.

ırics for Calculation

Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario		
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues		
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts		
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided		
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand		

Faculty In-charge	**Course Coordinator**	**Head of Department**
Signature: ______________	Signature: ______________	Signature: ______________
Date: ______________	Date: ______________	Date: ______________

a) Interpret the Scenario and identify [illegible]

The automated inspection system captures image of medical Components, but repetitive horizontal stripe Patterns appears due to electric interference. This type of degradation is Periodic noise, which is usually introduced by electrical or electro magnetic interference during image acquistion Periodic noise appears as repeated horizontal.

b) Explain How this degradation affects inspection accuracy.

Periodic noise reduces the visibility of small defects such Scratches, cracks, dents or Surface imperfections. Since the repetiti Stripe b/w defects and noise.

- False detection of defects
- Missing actual defects

c) Justify why your Selected approach is more Suitable metho

The frequency-domain approach is more effective Periodic noise is Concentrated at specific frequency locations. Filter Can Selectivity remove only these unwanted frequencies Preserving useful image details.

...ial - domain approach is more effective

...d cannot selectively remove periodic interference

...Recommend an appropriate enhancement technique to remove the

...interference

...Fourier Transform to convert the image into the frequency domain

...notch filter to remove the frequency components responsible for the periodic

...noise

...erse Fourier transform to reconstruct the enhanced image

...xplain the process in a logical sequence

...ge is captured from the inspection system

...rical interference introduces periodic horizontal stripes

... the Fourier transform to convert the image into the frequency

...

... the frequency peaks corresponding to the periodic noise.

... notch filter to remove these unwanted frequencies.

... the inverse Fourier transform to reconstruct the enhanced image.

...: Frequency-domain enhancement with a notch filter is the

... because it removes periodic noise effectively.

Autonomous Vehicle Navigation:

a) Interpret the Scenario and explain the challenge involved.

The Self-driving Car captures road images during heavy rain. Rain introduces random noise into the images, while important features Such as lane markings and road boundaries must remain clearly visible. The challenge is to remove the noise.

b) Identify the importance of Preserving edges while removing noise.

Edges Contain important structural information about the road environment.

- Detect the lane markings accurately
- Identify the road boundaries
- Recognize obstacles
- Make Safe steering and navigation.

) Recommend a Suitable filtering technique.

is most Suitable filtering technique is the Bilateral filter.

atial distance b/w pixels.

ensity difference b/w neighboring Pixels.

d)

Justify why your Selected filter preserves important image details

Unlike Gaussian or Mean filters, the bilateral filter avoids averaging across edges. it Smooths only pixels with Similar intensity values, thereby reducing noise while keeping lane markings and road edge sharp

This makes it highly Suitable for autonomous vehicle navigation more important than Complete Smoothing

) Present Your explanation Clearly and Systematically.

The vehicle Captures a road image during rainfall.

ain introduces random image noise.

ply a Bilateral filter to reduce noise.

e filter Smooths homogeneous regions while preserving edges.

ne markings and road boundaries remain sharp.

navigation System detects roads and obstacles more accurately.

driving decisions Can be made based on the enhanced image

lusion:- The bilateral filters Provides the best blance b/w noise

n and edge Preservation making it deal for autonomous vehicle

on in rainy Conditions.